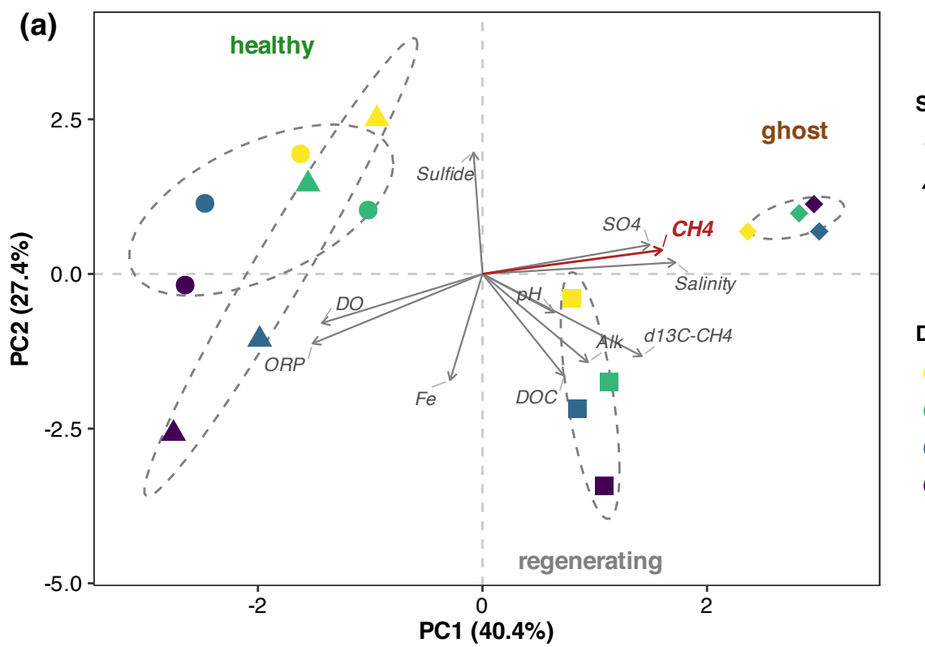
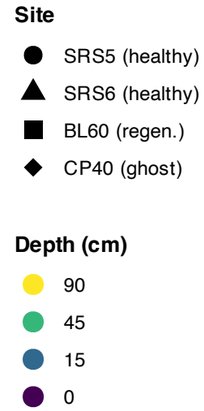
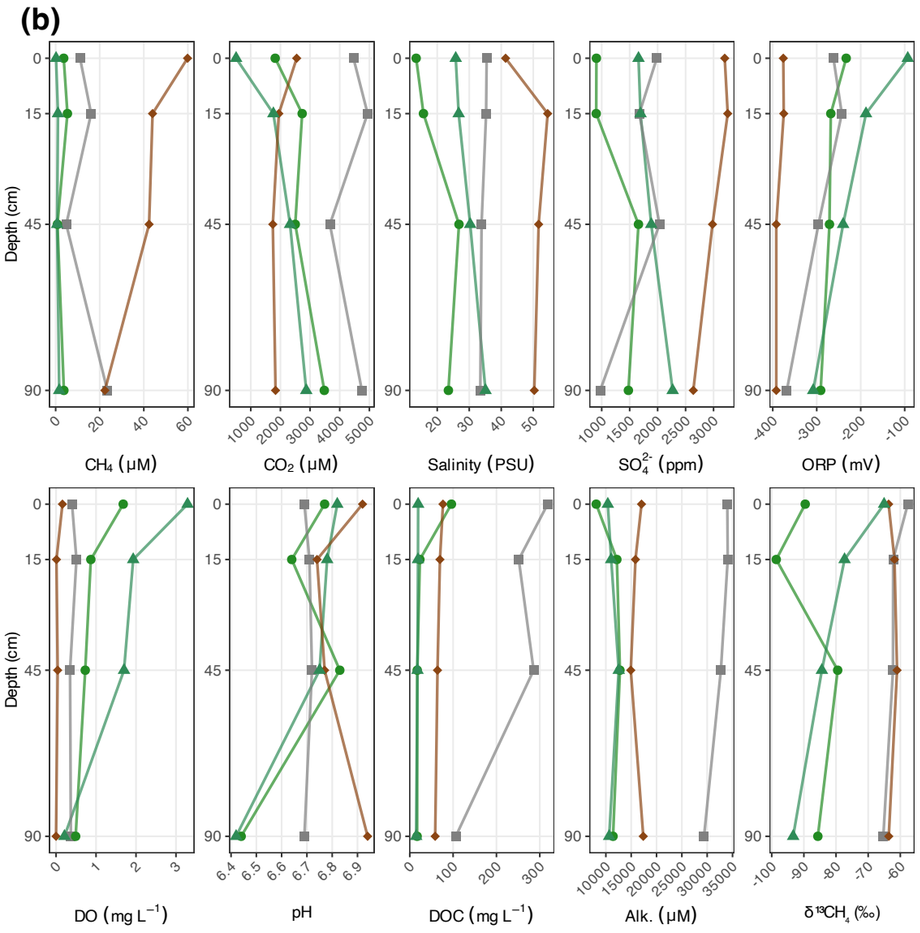
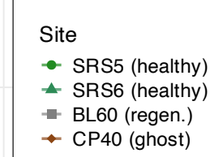
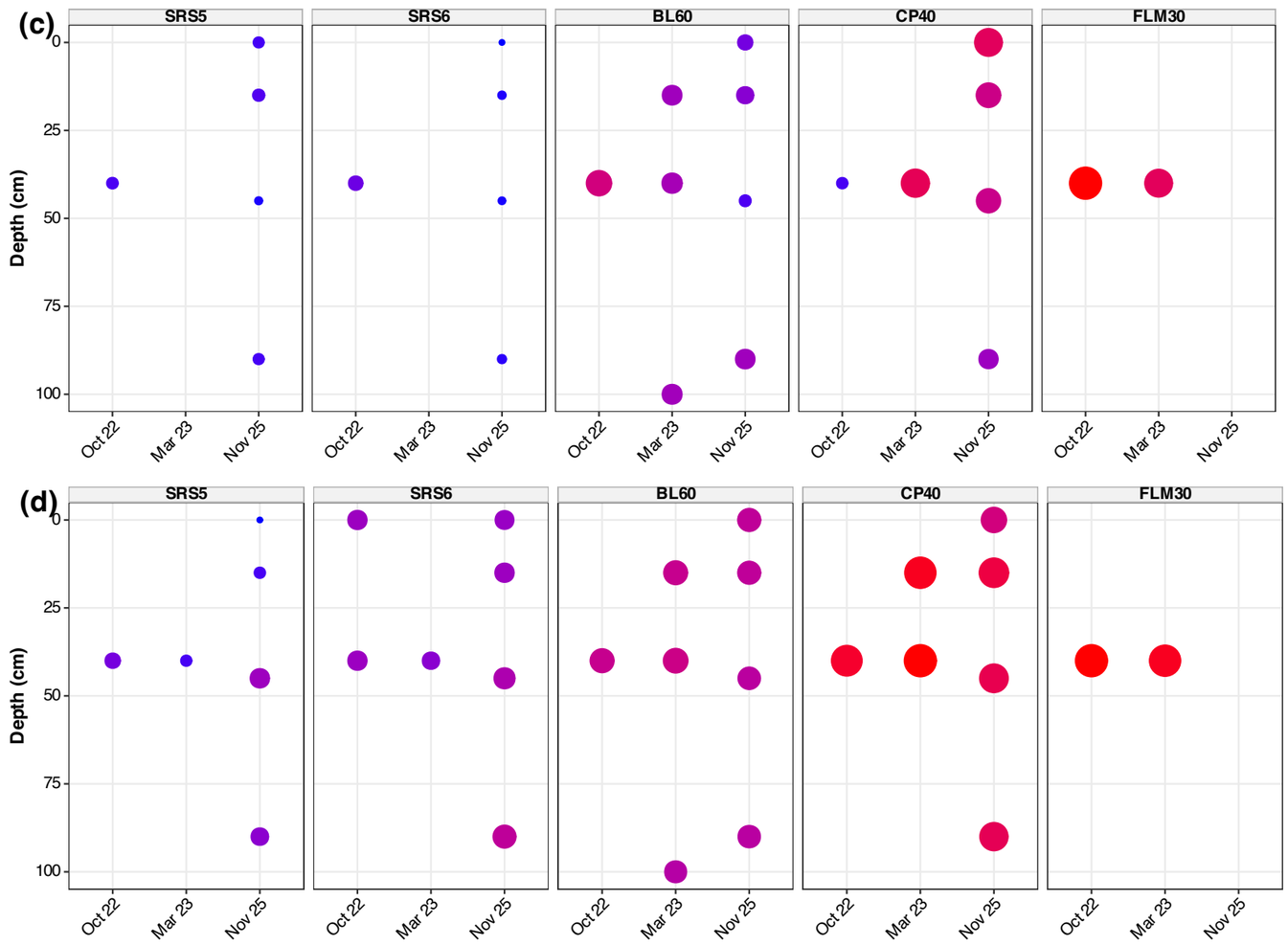

(a)
healthy
ghost
regenerating
Sulfide
SO4
CH4
Salinity
pH
DO
Alk
d13C-CH4
ORP
Fe
DOC
PC1 (40.4%)
PC2 (27.4%)

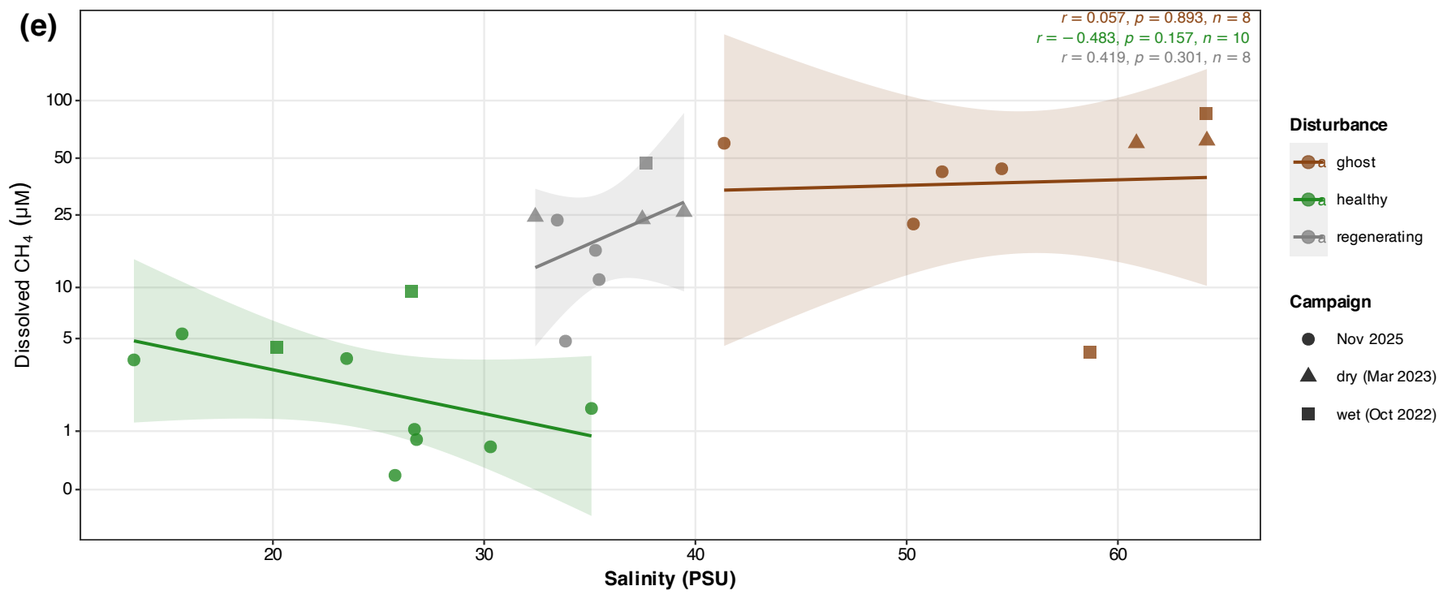

Site
SRS5 (healthy)
SRS6 (healthy)
BL60 (regen.)
CP40 (ghost)
Depth (cm)
90
45
15
0


(b)
Depth (cm)
CH4 (μM)
CO2 (μM)
Salinity (PSU)
SO4 2- (ppm)
ORP (mV)
DO (mg L−1)
pH
DOC (mg L−1)
Alk. (μM)
δ13CH4 (‰)


Site
SRS5 (healthy)
SRS6 (healthy)
BL60 (regen.)
CP40 (ghost)


(c)
SRS5
SRS6
BL60
CP40
FLM30
Depth (cm)
Oct 22
Mar 23
Nov 25
CH4 (μM)
PSU

(d)
SRS5
SRS6
BL60
CP40
FLM30
Depth (cm)
Oct 22
Mar 23
Nov 25
PSU


(e)
r = 0.057, p = 0.893, n = 8
r = −0.483, p = 0.157, n = 10
r = 0.419, p = 0.301, n = 8
Dissolved CH4 (μM)
Salinity (PSU)
Disturbance
ghost
healthy
regenerating
Campaign
Nov 2025
dry (Mar 2023)
wet (Oct 2022)